



IoT Christmas Tree

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Intro & Motivation

Project Scope:

- Geometry-based light animations
- Joystick-based robotics/animatronics
- Simple DIY setup for the Winter season
- An intersection of mechanical engineering and audio/visual design
- Valuable sensor inputs



Task breakdown

Dominic: LED control dashboard, brightness feedback, real-time plotting

Nathan: Animatronic design and fabrication, application control of servo motors, sensor feedback and autonomy

6 Layer IoT Model



06 Business

The users utilize the sensor data collected to optimize tree energy consumption



05 Application

Two html apps have the capabilities for LED and servo control and present collected data



04 Network

A Flask code runs from the RPi to execute the code which is brought online



03 Actuation

Sensor data and users control the LEDs and animatronic eyes



02 Data Processing

Sensor data is read by the Raspberry Pi, processed and used



01 Perception

Motion and brightness sensors interact with the physical world



Methods & Results

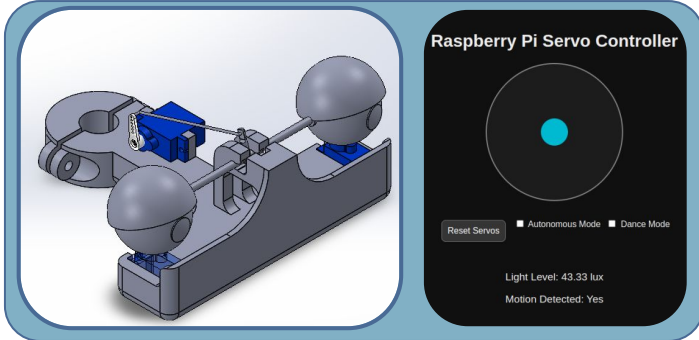


3D coordinates were collected for 50 WS2811 LEDs. Every animation uses these coordinates along with matrix transformations to generate light shows

BH1750 dims/amplifies the LEDs based on ambient brightness; publishes data on a plotly

Modular LED dashboard allows customizable light control and sensor feedback optimization (peak usage times)

The eyeball base plate was designed and manufactured to house 3 9g Servo motors. These Servo motors account for the individually moving eyes and the eyelid mechanism.



A joystick on the flask app controls the eyeballs in sync in addition to the eyelids.

BH1750 controls the opening and the closing of the eyelids in autonomous mode and reports it on the flask app

The motion detection is reported as well and should impact the motion of the eyes

Conclusions & Future Work

- 01** More user-friendly GUI design, maybe a mobile app
- 02** Music sync is difficult to implement due to latency and signal processing
- 03** Fine tune autonomous movements and sensor coordination
- 04** To increase resolution, 7-8 ft trees are next!

